Augmenting Behavioral and Business Science Surveys with Synthetic Noisy Data

Hohjin Im, PhD*
Independent Researcher, United States of America

Abstract

Imbalanced data poses a significant challenge in the behavioral and business sciences, often hindering the ability to derive meaningful and robust insights from niche or underrepresented segments. Although various methods have been developed to address this issue, each presents notable limitations that constrain their broader application in the field, e.g., traditional techniques like random oversampling risk data redundancy, synthetic respondent sampling may introduce confounding external biases, and synthetic minority oversampling tends to inflate spurious correlations. This paper introduces Synthetic Data Augmentation with Noise Injection (SYDNI), a novel approach that integrates noise injection with random forest models to address the limitations of traditional techniques such as SMOTE and GANs. SYDNI minimizes overfitting and spurious correlations while enhancing the robustness of statistical models. Using a case study of brand and product adoption among high-income consumers, this study demonstrates SYDNI's potential to augment imbalanced datasets and meet statistical power needs for niche subgroup analysis.

1 Introduction

Data imbalances are pervasive issues in academia for behavioral and business sciences, such as psychology, marketing, and organizational behavior. Recruiting an adequate sample size can be particularly challenging when the sample of interest is scarce in the population. For example, researchers studying electric vehicle (EV) consumers will face greater difficulty surveying owners of Lucid Motors vehicles than owners of Tesla vehicles because there are significantly fewer people buying the former than the latter (Kessel, 2024). Even in scenarios where the target group may be abundant in the general population, however, researchers may still encounter imbalanced datasets if there exist skews in the available participant pool. To illustrate, in psychology, convenience samples often come from university human subjects pools, where a disproportionate number of participants are women due to more women than men enrolling as psychology majors (Dickinson et al., 2012; J. Johnson et al., 2020). Consequently, many psychology study samples show a strong gender skew where the majority of participants are women.

Ideally, researchers would collect more data until the undersampled target (hereafter referred to as the *minority class*) reaches the desired sample size. However, in contrast to industry settings where funds for additional data collection may be more readily available, this is often neither feasible nor realistic for many academics facing various practical barriers, such as prohibitive costs and resource constraints. These challenges are particularly relevant for underfunded researchers, independent scholars, and students, who frequently lack access to the financial and infrastructural

*Correspondence: hohjini@uci.edu

resources available to their more privileged counterparts in the industry or well-funded academic institutions (Westfall & Yarkoni, 2016). Consequently, the expectation to meet costly field "best practices" disproportionately benefits well-funded institutions and concentrates high-impact publications within elite academic circles, marginalizing those who lack the necessary resources (Bergh et al., 2017; Larivière et al., 2015). This can perpetuate structural inequities in academic publishing and limit the diversity of perspectives contributing to the broader scientific discourse.

To address some of these barriers, I introduce SYDNI (Synthetic Data Augmentation with Noise Injection), a new synthetic oversampling algorithm as a practical avenue for researchers to generate high-quality synthetic data. Synthetic data can be used to conduct robustness checks and supplement analyses without the need for costly data collection. This method integrates traditional data augmentation techniques (e.g., random oversampling, noise injection) with machine learning (e.g., random forest) to produce robust synthetic data that closely resemble real observed data (Dimairo et al., 2015). I first provide brief introductions to common data balancing techniques already commonly used in the field and their utilities and limitations. I then introduce the new synthetic oversampling algorithm, provide an example use-case with evidence of its utility, and discuss its viability in behavioral and business research.

1.1 Data Balancing Techniques

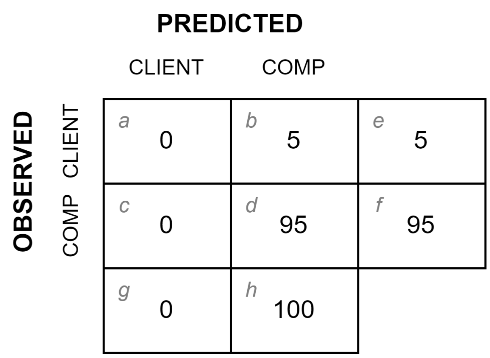

Figure 1: Confusion Matrix of Incorrect Classification with High Accuracy

Data balancing techniques have common use-cases where the research objective is to train and deploy models that classify an observation's membership in a group. Imbalanced data can lead to models favoring overrepresented groups (hereafter referred to as the *majority class*), resulting in subpar performance on the minority class and limited generalizability to novel, unseen data, as the models primarily learn to recognize patterns only exhibited in the majority class. This issue can be highly problematic in applied settings, where minority classes can often represent critical clusters for theoretical or practical insights. Consider, for example, a management study examining what leadership traits best prevent employee turnover. If only 5 out of 100 surveyed employees leave, the ratio of employee turnover to retention is 5:95 (Figure 1*e*-*f*). Consequently, a model trained on this data might default to predicting that no employee *ever* leaves (Figure 1*g*-*h*), yet still achieve 95% accuracy. This deceptively high accuracy, however, wholly fails to address the objective of understanding what drives employees to leave their organization.

Small sample sizes often limit researchers' ability to draw meaningful insights about minority classes in relation to the broader sample. Traditionally, sample weights have commonly been used to address the over- or under-representation of certain subgroups. However, this approach is inherently limited by its overemphasis of the minority class, which can amplify any existing biases and noise artifacts commonly present in smaller, more homogenous samples. More importantly, the use of sample weights is constrained to adjusting effects at the level of the total model. In other words, it fails to enable researchers to conduct subgroup analyses—treating the minority class as the sole sample of interest to conduct separate analyses on—which are critical for many initiatives, such as targeted clinical trials or strategy development. For example, a 5% occurrence rate of consumers buying private label products in a large sample of 2,000 consumers still only yields an $n = 100$ observations. Sample weights can amplify the effects of these 100 observations when analyzed at the total level, but the number of possible avenues of analysis significantly shrinks when needing to conduct analyses on *only* these customers. A range of techniques exists to generate artificially balanced data, each offering varying degrees of complexity depending on the context and objectives of the analysis.

1.1.1 Random Oversampling & Undersampling

At the simplest level, imbalances can be addressed by randomly undersampling the majority class or oversampling the minority class. Both techniques are computationally simple and do not require complex algorithms, making them common data preprocessing choices in classification problems with imbalanced data (Kotsiantis et al., 2006).

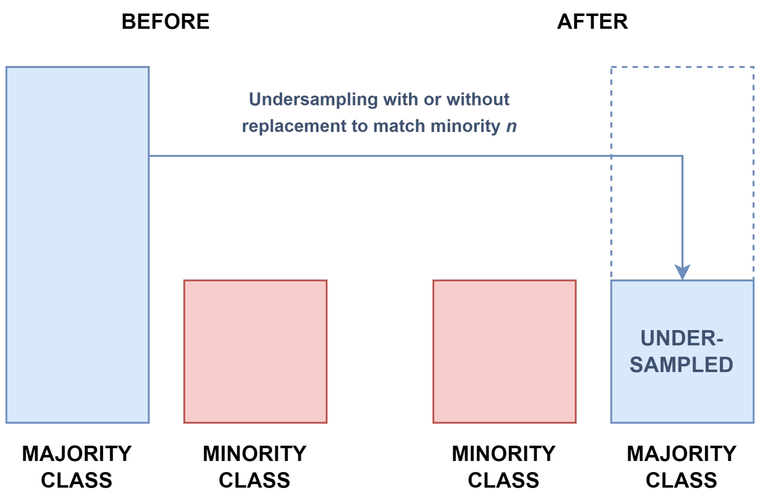

Figure 2: Random Undersampling

The former technique—*random undersampling* (RUS)—involves selecting a random subset of the majority class, either with or without replacement, until its sample size (n) matches that of the minority class or a specified n (Figure 2) (Japkowicz & Stephen, 2002; Kubat & Matwin, 1997). This approach reduces the risk of overfitting and enhances computational efficiency, which could be beneficial when the ratio of majority to minority class sizes is particularly pronounced (Drummond & Holte, 2003; Hasanin et al., 2019). However, RUS is limited by *information loss* as it eliminates potentially valuable data from the majority class. In disciplines like psychology, organizational behavior, and marketing, where datasets are often limited and nuanced, this loss can be detrimental by obscuring trends in the majority class and impairing the model's ability to detect and generalize meaningful patterns (Blagus & Lusa, 2013; Fernández et al., 2018; Van Hulse et al., 2007). For instance, in the example of employee turnover, undersampling the majority class could result in missing insights into broader reasons for employee retention.

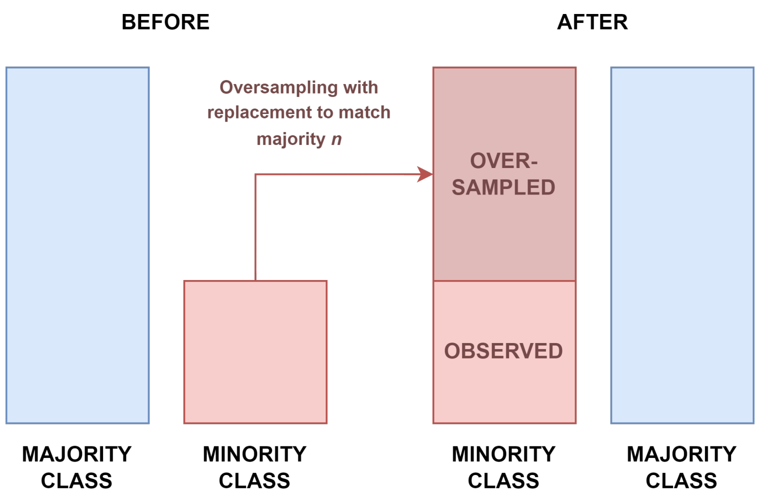

Figure 3: Random Oversampling

In contrast, the latter technique—*random oversampling (ROS)*—involves resampling a random observation of the minority class, with or without replacement, until its n matches that of the majority class or a specified n (Figure 3) (Japkowicz & Stephen, 2002; Kubat & Matwin, 1997). The advantage of ROS lies in its ability to retain the entirety of the observed data (Batista et al., 2004) while aiming to improve the detection of minority outcomes, like rare psychological phenomena or niche customer types (Thumpati & Zhang, 2023). However, ROS is limited by its propensity to induce model overfitting by introducing *data redundancy.* Data redundancy can cause the model to recognize duplicated data rather than true underlying patterns, especially in small or homogeneous datasets, thereby exacerbating noise detection (Buda et al., 2018; He & Garcia, 2009). For instance, oversampling may lead to an overemphasis on specific patterns relevant to only a niche group of unsatisfied employees, such as those that are overqualified for their roles, potentially skewing the understanding of why employees may seek employment elsewhere. Thus, although ROS can improve detection of minority class instances, it may simultaneously yield poor generalizability for the minority class.

1.2 Synthetic Data Generation

Synthetic data generation techniques provide alternatives to address the limitations of these traditional methods. Specifically, approaches like the *Synthetic Minority Over-Sampling Technique* (SMOTE), *Adaptive Synthetic Sampling* (ADASYN), and *Generative Adversarial Networks* (GANs) aim to produce synthetic datasets that retain core characteristics of the observed data. Although the concept of synthetic data generation has roots in the early days of computer science, its application to tabular survey data in the behavioral and business sciences is a more recent development.

1.2.1 Synthetic Minority Oversampling Technique (SMOTE)

SMOTE and ADASYN are two widely adopted techniques used to address class imbalance issues across various fields, such as medical diagnostics and fraud detection (Fernández et al., 2018; Fiore et al., 2019; Lemaıtre & Nogueira, 2017; López et al., 2013). SMOTE, developed by Chawla et al. (2002), operates by generating synthetic samples for the minority class (Figure 4). Rather than duplicating existing samples, SMOTE enhances minority class representation by creating new

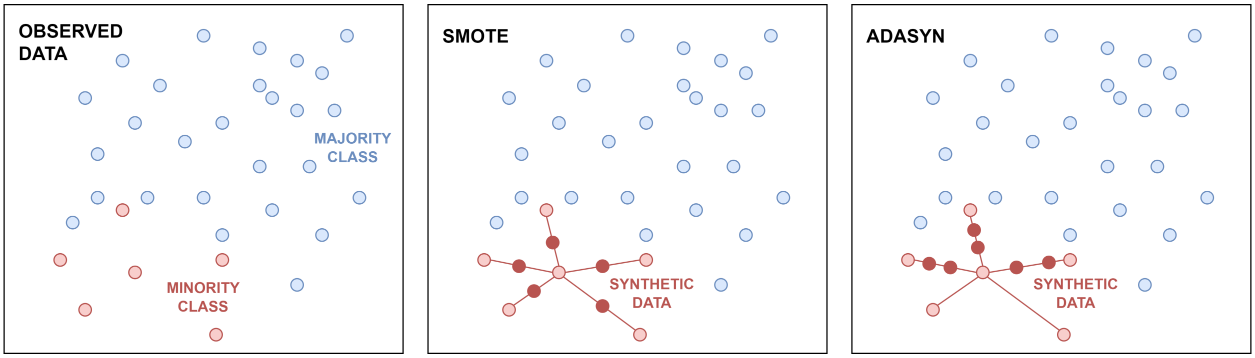

Figure 4: Synthetic Data Generation with SMOTE and ADASYN

data points that lie between existing minority observations (x) and their nearest neighbors (x_{nn}). That is, the process identifies the k-nearest neighbors for each minority instance and generates synthetic samples (x_{syn}) along the line segments connecting the original data point and its neighbors. This method relies on a random interpolation factor (λ), which takes a value between 0 and 1, ensuring the new synthetic data points are spread across the feature space [Chawla et al. (2002); @douzas2018]. Mathematically, this interpolation is represented as:

$$x_{syn} = x + \lambda(x_{nn} - x) \tag{1}$$

where x represents the original minority sample, x_{nn} is one of the nearest neighbors of x, and λ controls how far the new synthetic sample lies between these points. This interpolation process diversifies the dataset to reduce the risk of overfitting by distributing synthetic observations throughout the original data's feature space. However, SMOTE is susceptible to introducing information redundancy, particularly in regions where minority class observations are already abundant. This redundancy can lead to overfitting, where the model learns the synthetic noise rather than the true underlying patterns (Douzas & Bacao, 2018; Pradipta et al., 2021).

1.2.2 Adaptive Synthetic Sampling (ADASYN)

Building on the foundation of SMOTE, ADASYN, developed by He et al. (2008), takes a more focused approach to address class imbalance by prioritizing the generation of synthetic samples for minority class observations that are harder to classify, particularly those situated near decision boundaries, i.e., the line separating different classes in the feature space (Figure 4). Rather than treating all minority instances equally, ADASYN adapts the oversampling process by concentrating on the observations that are surrounded by majority class neighbors or are otherwise prone to misclassification. This focus allows the model to better handle challenging cases by refining the balance between classes, offering a more nuanced approach to oversampling (Gameng, 2019; He et al., 2008).

Like SMOTE, ADASYN generates new synthetic samples by interpolating between an original minority class observation (x_i) and one of its nearest neighbors (x_{zi}). The interpolation is guided by a random parameter (λ), which determines how far the synthetic sample lies along the line between the two points. Mathematically, this can be expressed as:

$$x_{syn} = x_i + \lambda(x_{zi} - x_i) \tag{2}$$

where x_{syn} is the newly generated synthetic sample, x_i is the original minority class observation,

x_{zi} is a randomly selected neighbor from the nearest neighbors of x_i, and λ is a random number between 0 and 1. By focusing on the more difficult-to-classify observations, ADASYN reduces bias towards the majority class and improves the model's performance in classifying minority instances. However, this emphasis on difficult cases can introduce noise near decision boundaries, which may lead to ambiguity and overlap between classes, complicating classification tasks (Buda et al., 2018; Garcia et al., 2010; He & Garcia, 2009; Thumpati & Zhang, 2023).

1.2.3 Generative Adversarial Networks (GANs)

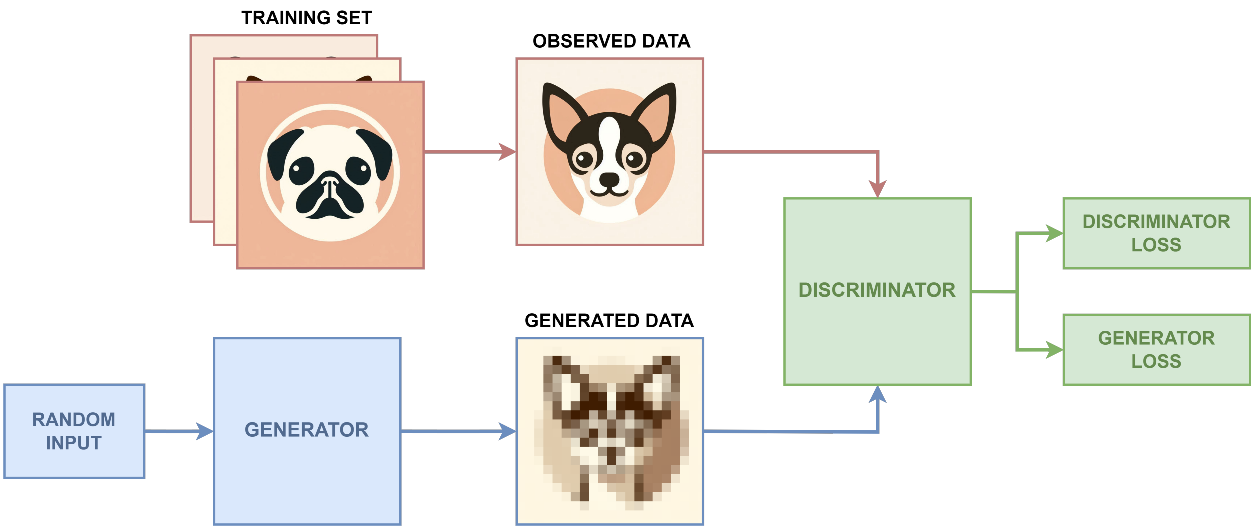

Figure 5: Generative Adversarial Networks

GANs, developed by Goodfellow et al. (2014), are widely recognized as a foundational framework in generative modeling due to their capacity in generating highly realistic synthetic data. GANs comprise two neural networks—a *generator* and a *discriminator*—that work in tandem: the Generator (G) creates synthetic data, while the Discriminator (D) attempts to distinguish between real and synthetic samples (Figure 5). Put simply, the Generator transforms random noise ($z \sim p_z(z)$) into synthetic data that resembles observed data, aiming to approximate the true data distribution $p_{\text{data}}(x)$. The Discriminator evaluates both observed ($x \sim p_{\text{data}}(x)$) and synthetic samples ($G(z)$), assigning a probability ($D(x)$) indicating whether a sample is real ($D(x) = 1$) or generated ($D(G(z)) = 0$). During training, both networks improve simultaneously through a competitive process. The Generator aims to *fool* the Discriminator by creating more realistic synthetic data, while the Discriminator aims to better distinguish between observed and synthetic samples. This process is formalized as a minimax game, where the objective function is:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))] \tag{3}$$

where $\mathbb{E}_{x \sim p_{data}(x)}$ represents the expected value over the observed data distribution, and $\mathbb{E}_{z \sim p_z(z)}$ represents the expected value over the noise distribution used to generate synthetic data. In other words, the Discriminator maximizes its ability to detect real data, while the Generator minimizes its detectability by producing more realistic synthetic data. This interaction allows the GAN to learn the characteristics of the real data and generate realistic synthetic samples (Goodfellow et al., 2014) without duplicating existing observed data, similar to SMOTE and ADASYN (Divovic et al., 2021; Zhu et al., 2017).

GANs have been successfully deployed in improving classification accuracy in highly imbalanced datasets, such as those in medical imaging and credit scoring (Vega-Márquez et al., 2020). Nonetheless, the application of GANs remains prohibitively complex for most researchers in behavioral and business sciences, where statistical training is predominantly focused on applied frequentist statistics and not in machine learning. Challenges to implementing GANs also include its propensity for instability during model training, in optimizing hyperparameters the complexity involved in supervised hyperparameter optimization, and the lack of definitive metrics for convergence often needed to prevent *mode collapse*, i.e., the generator fails to reproduce the full diversity of the training data, leading to outputs with limited variability (García-Esteban et al., 2023; Salimans et al., 2016). Further, their reliance on sophisticated architectures, computationally intensive processes, and rapidly advancing methodological developments demands a level of specialized expertise and infrastructure that is often inaccessible to non-specialists. These barriers make it unlikely for researchers outside of specialized machine learning fields to adopt GANs without significant interdisciplinary collaboration or additional training, particularly as resources needed to properly deploy GANs in practice may often not justify the incremental gains in data accuracy.

1.2.4 Spurious Correlations in Synthetic Data Generation

Despite the advantages of synthetic data generation over ROS and RUS, however, a significant limitation hindering its wider adoption in behavioral and business sciences is the issue of *spurious correlations*. SMOTE, ADASYN, and GANs all exhibit tendencies to inflate inter-item correlations, potentially leading to misleading interpretations of both the data and the models built upon it (Blagus & Lusa, 2013; Douzas & Bacao, 2018). Specifically, SMOTE and ADASYN's method of interpolating between existing data points or through local density estimations can amplify correlations, particularly in small or imbalanced datasets (Chawla et al., 2002; He et al., 2008). As the synthetic data is derived from neighboring observations, it often replicates existing patterns and artificially reinforces relationships that may not reflect the true distribution of the original data, thereby biasing models and distorting natural variability (Buda et al., 2018; Fernández et al., 2018).

Similarly, GANs, though designed to replicate overall data distributions, face issues such as *mode collapse*, where the generator produces a limited variety of samples and fails to capture the complexity and diversity of the dataset (Goodfellow et al., 2014; Salimans et al., 2016). This lack of variability can further exacerbate spurious correlations by generating overly similar observations to inflate dependencies between variables (Frid-Adar et al., 2018; Mariani et al., 2018). Additionally, interpolation mechanisms in SMOTE and ADASYN often introduce residual dependencies—a form of autocorrelation—when synthetic data points are generated close to observed ones, leading to dependent errors that skew model outcomes (Van Hulse et al., 2007). GANs, particularly when affected by mode collapse, may also produce overly structured samples, contributing to inflated residual dependencies.

1.3 Synthetic Data Augmentation with Noise Injection (SYDNI)

SYDNI was primarily developed in response to these limitations of existing data augmentation techniques in tabular human respondent data. SYDNI integrates ROS, noise injection, and random forest models to minimize risks of overfitting and emergence of spurious correlations that otherwise compromise the utility of synthetic data. By carefully balancing random noise with machine-learning-predicted values, SYDNI generates high-quality synthetic data that mirrors the statistical properties of the observed dataset, thereby improving the reliability and utility of the

Table 1: Comparison of Data Augmentation Techniques

Method	**Data Overfitting**	**Spurious Correlations**	**Complexity**	**Information Loss**	**Data Redundancy**
ROS	-	Low	Low	Low	High
RUS	-	Low	Low	High	Low
SMOTE	Medium	Medium	Medium	Low	Medium
ADASYN	Medium	Medium	Medium	Low	Medium
GANs	Low	High	High	Low	Low
SYDNI	Low	Low	Medium	Low	Low

data subsequent analyses.

As previously mentioned, existing synthetic data generation techniques frequently inflate inter-item associations to induce spurious correlations, raising Type I error rates and distorting the true associations present in the data. SYDNI addresses these challenges by introducing multiple layers of randomness into the synthetic data generation process. First, a proportion ($M\%$) of the observed data is randomly masked as missing, which is subsequently imputed using random forest models. This step serves a dual purpose of ensuring complete cases from existing missingness but also artificially injects missingness to reduce the chances of model overfitting. The imputed dataset is then random oversampled and injected noise to emulate the diversity that naturally arises from larger, more heterogeneous samples.

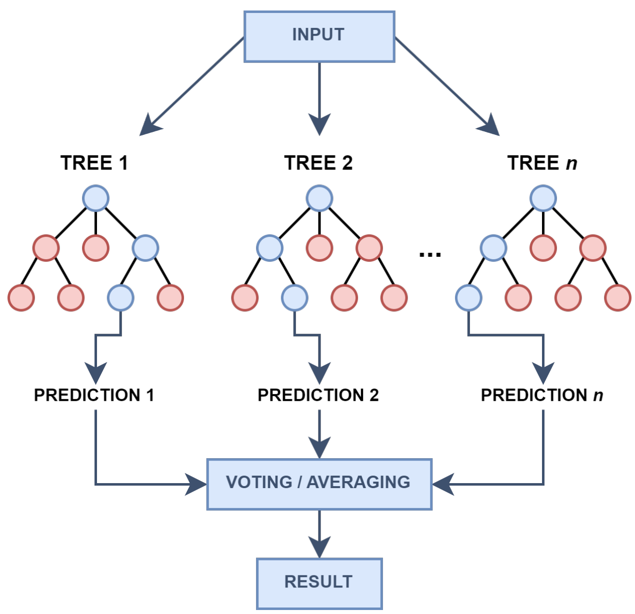

Figure 6: Random Forest Classification/Regression

SYDNI's reliance on random forests offers distinct advantages. Random Forests, introduced by Breiman (2001), operates by constructing an ensemble of multiple decision trees, each generated from a random subset of the data. The algorithm aggregates the predictions from all individual trees to produce a final prediction by voting (for classification) or averaging (for regression) the outputs (Figure 6). This ensemble approach harnesses the *wisdom of the crowd* to mitigate the risk of overfitting, thereby enhancing predictive accuracy and generalizability (A. Cutler & Zhao, 2001).

Further, random forest's ability to handle noise and missing values makes it highly effective in small-sample settings (D. R. Cutler et al., 2007), outperforming alternatives like Gradient Boosting Machines (GBM) and Artificial Neural Networks (ANNs). Although GBM and ANNs are powerful for certain applications, they are prone to overfitting in small datasets and demand extensive hyperparameter tuning and preprocessing, which can limit their practicality for synthetic data generation for academic research (Natekin & Knoll, 2013; Probst et al., 2019). Perhaps most importantly, random forest models excel in delivering high quality predictions at relatively low computational complexity compared to its other ML counterparts like GBM, ANNs, and GANs.

To further control spurious correlations, SYDNI implements a dual-structure approach in which only a fraction (X) of the final synthetic dataset is based on predicted values from the random forest models, while the remaining $(1-X)$ is derived from the noisy, oversampled data. This combination preserves the integrity of the learned associations while counteracting spurious correlations. The algorithm is iterated k times across different combinations of hyperparameters, selecting the final synthetic dataset based on how well it matches the observed data across key benchmark properties, such as means, standard deviations, inter-item correlations, skewness, and kurtosis. Lastly, SYDNI trains solely on the provided dataset, making it viable for sensitive data. By avoiding external data sources, SYDNI also mitigates the risk of introducing external biases while protecting the confidentiality of proprietary data.

2 Example Use-Case: Brand & Product Adoption Among High Income Consumers

Aside from data balancing techniques, sample weighting is a common alternative used in behavioral sciences to correct for under- or over-representation of certain clusters at the total sample. However, sample weighting is limited by both accessibility and statistical power. Accessibility issues stem from many publicly available tools (e.g., R packages) lacking sample weighting functionalities, where the availability of certain analytical features are often at the whims of the author. Proprietary software, however, often have numerous functionalities in response to user demand, but come at the price of high licensing costs, thereby excluding researchers or institutions with limited financial resources. Statistical power issues notably stem from sample weighting only adjusting for how much a minority class's effects influence the total model, but not improving the statistical power often needed for group comparisons. This issue becomes more pronounced when researchers need to filter their datasets to specific demographic clusters for analytical techniques that typically demand large sample sizes, such as multinomial logistic regression or structural equation modeling. Thus, SYDNI offers a more balanced solution to the issues associated with traditional ROS/RUS, synthetic data generation, and sample weighting.

The SYDNI algorithm has benefits in balancing minority with majority classes for robustness checks where researchers may be concerned that demographic or class skews may yield effect sizes not representative of the underlying population of interest. In other words, the SYDNI algorithm is also beneficial in use-cases where the research objective necessitates subgroup analyses for deriving important insights, yet the small sample sizes of certain subgroups inhibits the ability to confidently deem them robust. Subgroup analysis is a staple part of some fields, such as clinical and medical literature, but there is often a concern regarding whether the evident compelling effects are spurious (Sun et al., 2014).

To demonstrate the application of SYDNI, I present an example use-case in general consumer attitudes. Academic research that uncovers the generic attitudes, psychographics, and preferences underlying consumer behavior can be useful in providing important fundamental insights for strategizing the right sentiment in ad or product messages. Indeed, consumers' emotional states and cognitive frameworks play crucial roles in shaping their decision-making processes. Thus, the primary objective of this use-case study scenario is to identify how consumers' dispositional positive emotions are linked with one's cognitive value orientations, which, in turn, drive their general openness to adopt new products or brands.

2.1 Emotions & Values in Brand & Product Adoption

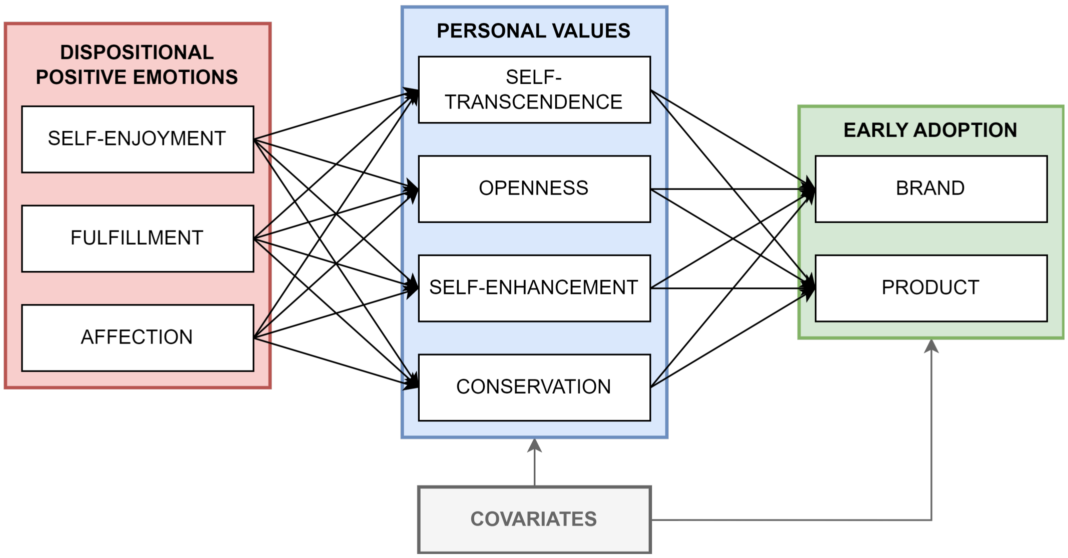

Figure 7: Explored Conceptual Path Model

Positive emotions like joy and pride expand individuals' cognitive and behavioral repertoires, fostering an openness to new experiences and personal growth (Fredrickson, 2001) which emphasizes self-direction and stimulation (Schwartz, 1992). Consumers who frequently experience positive emotions are more likely to adopt values prioritizing exploration, creativity, and novelty (Verplanken et al., 2009). Bettiga & Lamberti (2018), for example, demonstrated that anticipated emotions, such as excitement, significantly drive both the initial adoption of products and their continued use. Thus, those experiencing elevated propensity for positive emotions may promote an intrinsic motivation to seek new experiences, including adopting new products and brands (Bagozzi et al., 1999; Lavigne et al., 2011).

Values, on the other hand, may serve as cognitive filters to shape how emotions are processed and aligned with behavior. For example, individuals with strong openness to change values are more likely to interpret positive emotions as motivation to seek new experiences and adopt novel products (Bilsky & Schwartz, 2008; Wang et al., 2008). In contrast, those with strong conservation values, which prioritize stability and tradition, may interpret the same emotions as potential threats to their preference for security, thereby dampening the likelihood of adopting new products despite experiencing positive emotions (Bettiga & Lamberti, 2018; Wang et al., 2008). Thus, I examine a path model with dispositional positive emotions as the predictors and personal values as mediators to brand and product adoption (Figure 7).

2.2 Application of SYDNI to Augment High Income Consumers

For the purposes of the current illustration, the product of interest in question for further marketing campaign initiatives can be inferred to be of interest to middle and high income consumers. Thus, the objective is for the model to represent the associations of emotions and values with brand and product adoption among only middle and high income consumers (i.e., the opportunity consumer segments). In doing so, the dataset being used to estimate the effects should, ideally, represent the consumer base of interest. Without a specific product line to specify a niche consumer base of interest, the model can be based on the general consumer population. The problem, however, is that the current datafile does not yield a representative distribution of middle to high income classes. In the MM dataset, there are 801 respondents from middle-income households (50,000-149,999 USD annual household income [HHI]) and 123 respondents from high-income households (150,000+ USD HHI). In contrast, the U.S. population exhibits approximately a 2:1 ratio of middle to high HHI (US Census 2023). Thus, high-income consumers are notably underrepresented, with a shortfall of 277 respondents from the necessary 400 to reflect the general population ratio. Consequently, models using this sample distribution will be skewed toward middle-income consumers due to the underrepresentation of high-income individuals.

3 Methods

3.1 Participants & Procedure

A total of 2492 respondents were sampled for the Measuring Morality project study (K. M. Johnson et al., 2014; Vaisey & Miles, 2014), of which a nationally representative sample of 1519 respondents participated (61% cooperation rate). The Measuring Morality project was conducted in March 2012 by GfK North American on behalf of Duke University. The study consisted of various morality psychological scales and demographic questions designed to understand the various domains of morality. All respondents were compensated $5 for their participation.

3.2 Measures

Table 2: Measurement Reliability & Validity of Scales

Composite	α	ρ_c	**AVE**	$\lambda \in$	$\mathbf{W} \in$
Personal Values					
Self-Transcendence	0.623	0.841	0.726	[0.851, 0.853]	[1.000, 1.000]
Openness	0.720	0.848	0.652	[0.697, 0.879]	[0.882, 1.124]
Self-Enhancement	0.782	0.903	0.822	[0.897, 0.917]	[1.000, 1.000]
Conservatism	0.751	0.858	0.669	[0.787, 0.846]	[0.955, 1.031]
Disp. Positive Emo.					
Self-Enjoyment	0.783	0.902	0.822	[0.905, 0.908]	[1.000, 1.000]
Fulfillment	0.747	0.860	0.673	[0.736, 0.884]	[0.886, 1.095]
Affection	0.790	0.905	0.827	[0.904, 0.915]	[1.000, 1.000]

Note:
α = Cronbach's alpha; ρ_c = Composite Reliability; AVE = Average Variance Extracted; λ = Loading; W = Weight.

Measurement reliabilities are given in Table 2. *Dispositional positive emotions* were measured using the 21-item version of the Dispositional Positive Emotions Scale (DPES). Three higher-order composite emotions of Self-Enjoyment (Joy, Pride), Fulfillment (Contentment, Amusement, Awe), and Affection (Love, Compassion) were calculated. *Personal cognitive values* were measured using the 21-item Schwartz Values scale consisting of ten values along two bipolar dimensions: Openness to Change (Self-Direction, Hedonism, Stimulation) vs Conservation (Security, Conformity, Tradition) and Self-Enhancement (Achievement, Power) vs. Self-Transcendence (Universalism, Benevolence). New brand and product adoption tendencies were both measured using single items (Brand: "I often try new brands because I like variety"; Product: "I am usually the first of my friends to try new products and services").

4 Results

Synthetic data were generated using five different methods: SYDNI, Synthpop, SMOTE, ADASYN, and GANs. To achieve the target of 400 high-income respondents (inclusive of 123 observed respondents), SYDNI was used to generate 277 synthetic observations trained on the 123 observed high-income respondents, using 100 random forest decision trees. The proportion of missingness randomly introduced into the data prior to model training was set at 10%, 20%, and 30%. The proportion of noisy oversampled data replaced with predicted values from the RF model was set at 30%, 50%, and 70%. All combinations of missingness and replacement parameters were iterated 10 times, resulting in 90 unique iterations of synthetic datasets. The quality of the synthetic data was evaluated using a weighted scoring scheme. Variable means and standard deviations were each assigned a weight of 20%, based on how closely the synthetic data aligned with the observed means and standard deviations. The correlation index, weighted at 40%, evaluated the similarity between synthetic and observed inter-item correlations, with a Type II error correction applied to penalize synthetic datasets exhibiting spurious correlations relative to the observed data. Skewness and kurtosis were each weighted at 10%, with bias correction enabled to prioritize synthetic data approximating a normal distribution. The synthetic dataset with the highest global index score was selected for further analysis, while other iterations were excluded from consideration. A beta release of the tool can be found as a Shiny dashboard app here (https://imh-ps.shinyapps.io/SYDNI/). Alternatively, the `R` syntax used to run SYDNI with these settings is given as follows:

```r
# Install the `obms` package
devtools::install_github("imh-ds/obms")

# Load in the `obms` package
library(obms)

# Generate augmented data using the `sydni` function
sydni_augment <- sydni(
  data = high_income_data,       # Specify dataframe of subgroup to train SYDNI on
  covariates = covariates,       # Specify vector of covariates
  tree_n = 100,                  # Set n trees to 100
  aug_n = 277,                   # Set augmented sample size to 277
  train_prop = c(.1, .2, .3),    # Training NA injection set to 10%, 20%, 30%
  aug_prop = c(.3, .5, .7),      # Augmentation replacement set to 30%, 50%, 70%
  noise_value = 1,               # Noise value of -1 to +1
```

```
  iter_n = 10,                    # Iterate SYDNI 10 times
  mean_index_weight = 0.2,        # Mean index weight set to 20%
  sd_index_weight = 0.2,          # SD index weight set to 20%
  cor_index_weight = 0.4,         # Correlation index weight set to 40%
  skew_index_weight = 0.1,        # Skew index weight set to 10%
  kurtosis_index_weight = 0.1, # Kurtosis index weight set to 10%
  error_correction = TRUE,        # Type II error penalization
  bias_correction = TRUE,         # Normal distribution prioritization
  return_all_mods = TRUE          # Return all augmentation iterations
)

# Check model performances
sydni_augment$accuracy_table # Model 70 considered best performance

# Save Model 70 synthetic data as augmented dataframe
augmented_data <- sydni_augment$augmented_data$model_70$aug_data
```

Table 3: Measurement Reliability & Validity of Scales of Augmented Data

Composite	α	ρ_c	**AVE**	$\lambda \in$	**W** $\in$
Personal Values					
Self-Transcendence	0.684	0.864	0.761	[0.856, 0.889]	[1.000, 1.000]
Openness	0.812	0.891	0.732	[0.801, 0.915]	[0.945, 1.094]
Self-Enhancement	0.835	0.926	0.862	[0.916, 0.941]	[1.000, 1.000]
Conservatism	0.781	0.874	0.697	[0.812, 0.858]	[0.964, 1.044]
Disp. Positive Emo.					
Self-Positivity	0.784	0.903	0.823	[0.904, 0.911]	[1.000, 1.000]
Fulfillment	0.803	0.889	0.729	[0.795, 0.934]	[0.938, 1.123]
Affection	0.859	0.935	0.877	[0.932, 0.941]	[1.000, 1.000]

Note:
α = Cronbach's alpha; ρ_c = Composite Reliability; AVE = Average Variance Extracted; λ = Loading; W = Weight.

Table 3 provides the measurement reliability and validity statistics of the study measures using the SYDNI augmented data. Compared to the observed total data in Table 1, the SYDNI augmented data shows comparable, but slightly higher, composite reliability and AVE, which can be attributable to higher item loadings.

Table 4 provides the means comparisons of the observed and augmented high-income respondents. Across all covariates and key variables of interest, the augmented data showed no statistically significant difference with the observed data. Only Self-Transcendence showed a small effect size difference with $g = 0.155$ with no statistical significance.

SynthPop was implemented using the `synthpop v.1.8-0` R package (Nowok et al., 2016) to generate fully synthetic data using classification and regression trees (CART). Originally designed to create synthetic datasets that replace sensitive information while preserving the statistical properties of the original data, SynthPop synthesizes variables one at a time, drawing synthetic values based on previously synthesized variables. Although synthpop and SYDNI both rely on tree-based

Table 4: Mean Differences Between Observed and Augmented Data

Outcome	**Group**	***n***	**Mean**	**SD**	***df***	***t***	***p***	***g***
Female	Augmented	277	0.563	0.497	232.671	0.341	0.733	0.037
	Observed	123	0.545	0.500				
Age	Augmented	277	49.141	13.777	213.502	0.633	0.527	0.071
	Observed	123	48.122	15.294				
Education	Augmented	277	11.903	1.664	236.154	0.137	0.891	0.015
	Observed	123	11.878	1.648				
Paycheck to Paycheck	Augmented	277	1.805	0.879	237.170	0.347	0.729	0.037
	Observed	123	1.772	0.867				
Self-Image	Augmented	277	3.386	0.642	231.744	0.756	0.451	0.082
	Observed	123	3.333	0.649				
Self-Transcendence	Augmented	277	4.415	0.674	192.593	-1.315	0.190	0.155
	Observed	123	4.530	0.853				
Openness	Augmented	277	3.782	0.797	200.698	-0.326	0.745	0.038
	Observed	123	3.814	0.955				
Self-Enhancement	Augmented	277	3.468	0.874	197.993	-0.037	0.971	0.004
	Observed	123	3.472	1.065				
Conservation	Augmented	277	3.819	0.771	192.623	-0.210	0.834	0.025
	Observed	123	3.840	0.975				
Self-Enjoyment	Augmented	277	5.481	0.596	201.366	-0.819	0.414	0.095
	Observed	123	5.541	0.711				
Fulfillment	Augmented	277	4.997	0.833	216.622	-0.587	0.558	0.066
	Observed	123	5.054	0.909				
Affection	Augmented	277	5.214	0.815	196.677	-0.815	0.416	0.095
	Observed	123	5.298	1.002				
Brand Adoption	Augmented	277	2.913	0.880	191.249	0.842	0.401	0.100
	Observed	123	2.817	1.124				
Product Adoption	Augmented	277	2.076	0.652	204.309	-0.069	0.945	0.008
	Observed	123	2.081	0.764				

models—where SynthPop uses CART and SYDNI uses random forest—SYDNI introduces a key difference by retaining a portion of noisy oversampled responses in the final synthetic data, which distinguishes it from synthpop's fully synthesized output. Thus, SynthPop is also tested as a close sister alternative to SYDNI.

Both SMOTE and ADASYN were implemented using the `smotefamily v.1.4.0` R package (Siriseriwan, 2024). Both techniques were conducted conducted with $k = 5$ nearest neighbors and a duplication size of 3. This duplication generated 369 synthetic observations, from which 277 were randomly selected to match the desired augmentation sample size of the high-income group. The resulting synthetic data was then rounded to maintain the consistency of observed numeric values. GANs synthetic data were generated using the `RGAN v.0.1.1` R package (Neunhoeffer, 2022) without specific parameter tuning to demonstrate GANs' limited applicability in behavioral survey data without extensive model optimization. The data were standardized prior to training, and a random noise vector is generated as input to feed into the generator. GANs training was conducted using dropout regularization, with 277 synthetic samples generated. Two models were trained: one with dropout and one without. Dropout regularization was ultimately employed to mitigate overfitting and improve the generalization of the model. The synthesized high-income data was then transformed back to the original scale and rounded to ensure comparability with the observed data.

Table 5: Correlation with Brand Adoption by Synthetic Data Generation Technique

Variable	Observed	SYDNI	Synthpop	GANs	SMOTE	ADASYN
Self-Transcendence	0.084	0.034	0.143*	−0.136*	0.214***	0.179**
Openness	0.319***	0.300***	0.241***	0.089	0.405***	0.257***
Self-Enhancement	0.169	0.160**	0.072	0.152*	0.238***	0.127*
Conservation	0.030	0.031	0.067	−0.089	0.139*	0.140*
Self-Enjoyment	0.212*	0.149*	0.176**	−0.077	0.254***	0.133*
Fulfillment	0.046	0.006	0.086	−0.129*	0.122*	0.068
Affection	0.103	0.070	0.068	0.280***	0.267***	0.136*

Table 5 provides the bivariate correlations between Brand Adoption and the 3 dispositional positive emotions and 4 cognitive values for the observed high-income consumers and the synthetic high-income consumers generated via SYDNI, Synthpop, GANs, SMOTE, and ADASYN. SYDNI data generally maintain the directional patterns of the observed correlations with minor correlation penalizations. Synthpop data also performs well but suffers from an inflated effect from Self-Transcendence and deflated effect from Self-Enhancement. On the other hand, ADASYN data yields inflated effects from Self-Transcendence and Conservation. SMOTE data yielded inflated effects across all variables and GANs data yielded both inflated and reverse effects.

Table 6: Correlation with Product Adoption by Synthetic Data Generation Technique

Variable	Observed	SYDNI	Synthpop	GANs	SMOTE	ADASYN
Self-Transcendence	−0.005	0.067	0.013	−0.349***	0.125*	0.023
Openness	0.337***	0.284***	0.143*	−0.033	0.407***	0.309***
Self-Enhancement	0.202*	0.208***	−0.014	0.022	0.332***	0.249***
Conservation	−0.018	0.022	0.009	−0.174**	0.160**	−0.030
Self-Enjoyment	0.224*	0.215***	0.076	0.154*	0.301***	0.194**
Fulfillment	0.049	0.066	0.018	−0.087	0.081	−0.057
Affection	0.046	0.079	−0.080	0.125*	0.198***	0.010

Table 6 provides the bivariate correlations for Product Adoption. Like in the case of Brand Adoption (Table 5), SYDNI data generally maintain the directional patterns of the observed correlations with minor correlation penalizations. ADASYN performs notably better for Product Adoption by retaining the general patterns of effects with similar magnitudes. However, Synthpop data falter by yielding deflated effect sizes for Openness, Self-Enhancement, and Self-Enjoyment. GANs data yields inflated effects for Self-Transcendence, Conservation, and Affection while yielding deflated effects for Openness and Self-Enhancement. SMOTE data again show inflated effects across all variables.

Table 7: Classification of Synthetic vs. Observed Data

Metric	**SYDNI**	**SynthPop**	**GANs**	**SMOTE**	**ADASYN**
Accuracy	0.521	0.481	0.982	0.424	0.448
Precision	0.520	0.480	0.990	0.429	0.451
Recall	0.585	0.463	0.973	0.471	0.498
F1 Score	0.549	0.469	0.981	0.447	0.472

Each model's synthetic data were appended to the real observed data for classification analysis. To ensure balanced class representation, the synthetic observations in each augmented dataset were randomly undersampled to match the sample size of the real observed data, achieving a 1:1 ratio of synthetic to real observations. Random forest classification was conducted 100 times for each model, aiming to distinguish synthetic data from real observations, and the reported metrics represent averages across these iterations. Model performance was benchmarked against a hypothetical random classifier, for which Accuracy, Precision, Recall, and F1 Score would be 0.5, indicating complete indistinguishability between synthetic and real data (i.e., predictability akin to random guessing). Table 7 summarizes these results.

The findings reveal that SynthPop and SYDNI perform comparably to a random classifier, with metrics consistently near 0.5, suggesting minimal distinguishability between synthetic and real data. Conversely, GANs display high predictive performance, with metrics deviating substantially from 0.5, indicating clear distinguishability between the two classes. SMOTE and ADASYN exhibit systematic misclassification biases, with lower metrics overall, though ADASYN's Recall and F1 Score are slightly closer to 0.5 than those of SMOTE. Overall, SynthPop and SYDNI emerge as the most effective methods for generating synthetic data that are indistinguishable from real observations.

Table 8 provides the PLS-SEM direct effects for the model inclusive of both observed and SYDNI-augmented data. Self-Enjoyment and Fulfillment were positive correlates of Openness. Affection was a strong positive correlate of Self-Transcendence. In turn, Openness was the biggest positive driver of both brand and product adoption while Self-Transcendence was a negative driver of brand adoption. Accordingly, Self-Transcendence mediated the link between Affection and Brand Adoption ($\beta =$ -0.057, SE = 0.018, $z =$ -3.189, 95% CI [-0.091, -0.024], $p = 0.001$). Openness likewise mediated the effects of Self-Enjoyment ($\beta = 0.119$, SE = 0.018, $z = 6.470$, 95% CI [0.086, 0.156], $p < 0.001$) and Fulfillment ($\beta = 0.037$, SE = 0.012, $z = 2.984$, 95% CI [0.015, 0.062], $p = 0.003$) on Brand Adoption. Similarly, Openness mediated the effects of Self-Enjoyment ($\beta = 0.079$, SE = 0.017, $z = 4.720$, 95% CI [0.048, 0.111], $p < 0.001$) and Fulfillment ($\beta = 0.024$, SE = 0.009, $z = 2.768$, 95% CI [0.010, 0.043], $p = 0.006$) on Product Adoption.

These effects are largely consistent with the effects seen in the model using only observed data, with only minor changes in the size of effects but not directionality (see Appendix A1). This aligns with and illustrates the primary function of SYDNI—to augment existing data and stabilize model effects for more robust inferences reflective of the underlying population of interest (see Appendix Tables A2 and A3 for subgroup differences).

Table 8: Direct Effects of Observed + Augmented Opportunity Income Consumers

	Model 1 Transcendence	Model 2 Openness	Model 3 Enhancement	Model 4 Conservation	Model 5 Brand	Model 6 Product
Variable	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*
self	-0.008	0.371 ***	0.455 ***	0.121 **	0.066	0.072
	(-0.080, 0.067)	(0.303, 0.440)	(0.384, 0.521)	(0.035, 0.198)	(-0.016, 0.147)	(-0.009, 0.162)
fulfillment	0.011	0.115 **	-0.099 **	-0.059	-0.050	0.044
	(-0.050, 0.076)	(0.048, 0.184)	(-0.174, -0.024)	(-0.140, 0.017)	(-0.132, 0.028)	(-0.037, 0.123)
affection	0.468 ***	-0.026	-0.050	0.235 ***	0.103 *	0.007
	(0.400, 0.532)	(-0.106, 0.050)	(-0.127, 0.028)	(0.150, 0.318)	(0.017, 0.190)	(-0.084, 0.096)
transcendence					-0.122 **	-0.054
					(-0.195, -0.049)	(-0.127, 0.016)
openness					0.321 ***	0.212 ***
					(0.246, 0.398)	(0.134, 0.288)
enhancement					-0.039	-0.036
					(-0.117, 0.040)	(-0.115, 0.046)
conservation					0.006	-0.005
					(-0.065, 0.077)	(-0.073, 0.060)
Observations	1201	1201	1201	1201	1201	1201
R^2	0.241	0.225	0.248	0.155	0.110	0.100
R^2 Adj	0.235	0.219	0.242	0.149	0.100	0.090
F	42.031	38.496	43.568	24.283	11.245	10.152

Note:
Demographic variables and other covariates not shown.

5 Discussion

The widespread use of machine learning in industry has brought advancements in data balancing techniques to the forefront of applied research. However, despite their commonality in applied settings, these techniques remain relatively underutilized and overlooked in academic research, particularly within the behavioral sciences where tabular survey data are common. This gap between industry and academia represents a missed opportunity, as data augmentation methods, when properly applied, hold considerable potential for improving the robustness and generalizability of findings. As demonstrated in the present use-case, the SYDNI algorithm provides a valuable tool for addressing data imbalances and small subgroup sample sizes in a manner that both supplements traditional analyses and offers a pathway for performing robustness checks. By leveraging SYDNI, researchers can mitigate the limitations of small or imbalanced datasets, ensuring that their findings are both statistically reliable and more representative of broader population distributions. SYDNI's potential to bridge this gap highlights the untapped utility of data balancing methods in academic research, opening the door for more refined and scalable analyses that have long been a staple in industry but has been wholly absent in academia.

5.1 SYDNI Utilities

The SYDNI algorithm is primarily designed to perform robustness checks by augmenting existing datasets rather than to synthesize entire new datasets. When novel data is required, alternative approaches, such as synthetic respondent sampling via large language models (LLMs) trained on different parameters, may be another viable alternative approach. SYDNI's main utility lies in its capacity to augment primary analyses by determining whether observed effects within specific subgroups or across the entire dataset remain robust after accounting for representativeness and introducing penalties that emulate the increased data heterogeneity typically observed as sample sizes grow (Probst et al., 2019). As demonstrated in the example use-case, SYDNI can be used to augment a minority subgroup to the desired ratio relative to the majority class and see little shift in the broader model effects. This illustrates that the observed model effects are not a byproduct of biases from the majority group and is robust even when the sample distributions are changed to reflect the underlying population of interest.

SYDNI proves most useful in cases where small sample sizes limit researchers' ability to conduct thorough analyses or draw reliable inferences. Although the growing insistence from academic journals, editors, and reviewers on large sample sizes is driven by the valid goal of reducing Type II errors, it risks marginalizing both research topics where recruitment of large samples is difficult (e.g., studies involving undocumented immigrants or prisoners) and researchers who lack institutional or financial resources to easily collect more data (e.g., independent researchers, students, underfunded institutions). SYDNI therefore serves as a potential tool for such cases, making the publication of robust research results more accessible by offering alternative solutions to addressing small sample challenges commonly faced by those outside well-funded academic circles.

5.2 SYDNI Considerations

Nevertheless, SYDNI is not intended to artificially inflate sample sizes to extreme levels (e.g., from $n = 50$ to $n = 10{,}000$) for the sole purpose of achieving statistical significance, as such practices would fall into questionable research conduct, such as p-hacking (Nuzzo, 2014). In instances where significant sample augmentation is needed to achieve desired ratios for certain subgroups, researchers should prioritize effect sizes over statistical significance. This is especially relevant given the sensitivity of p-values to sample size, raising the potential for SYDNI to be misused for p-hacking, whether intentionally or not. Effect size interpretations offer a more reliable alternative when samples are scaled appropriately to ensure sufficient statistical power (Cumming, 2013).

Consider, for instance, a model run on a small subgroup with $n = 100$, where a marginal association between variables X and Y reveals an effect size of $r = 0.2$ and a p-value of 0.049. If this model is underpowered and lacks robustness, relying on such results for broad claims would be questionable. However, if an effect size of $r = 0.15$ is deemed practically significant for the study's topic and field, SYDNI could augment the sample size to stabilize the effects. By increasing the total sample to $n = 300$ and applying penalties to reduce inter-item correlations to simulate the natural increase in data variability stemming from increased sample heterogeneity, SYDNI helps assess whether the observed effect is robust to noise. If, in the augmented dataset, the effect size drops to $r = 0.1$, researchers could reasonably conclude that the original association in the observed data was likely unstable or influenced by outliers, rather than indicating a meaningful effect, regardless of whether the new p-value is statistically significant or not.

Additionally, as SYDNI is trained solely on observed data, it does not incorporate models trained on external data (such as LLMs). Consequently, the augmented data will inherently reflect some of the inherent biases present in the original sample, regardless of the noise injected during model training. SYDNI is designed to oversample the existing data, simulating what the sample would likely look like if additional respondents were drawn from the same population. For instance, in an employee experience (EX) study with 500 female and 300 male public administration employees, SYDNI could augment 200 male employees to achieve a balanced gender ratio. These 200 synthetic employees would mimic the characteristics of real public administration employees, but would not be representative of employees from other organizations, such as Google or Microsoft. Thus, SYDNI's synthetic data is intended to extend the scope of the original sample, not to generalize beyond the population from which the sample was drawn. Sampling limitations of the observed data will therefore remain relevant even with the application of SYDNI.

5.3 Integration With Synthetic Respondent Sampling

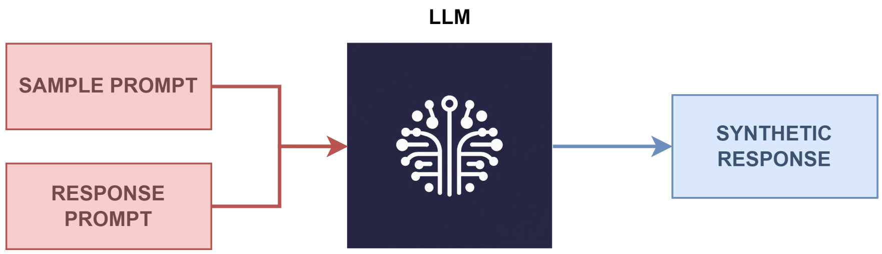

Figure 8: Synthetic Sampling with LLMs

SYDNI may, however, yield greater utility in the future. Recent advancements in generative artificial intelligence (genAI) and LLMs (e.g., ChatGPT, Claude) have introduced *synthetic respondent sampling* as a novel alternative to data augmentation by eliminating the need to collect primary training data altogether. LLMs are already trained on extensive datasets using transformer architectures (OpenAI et al., 2024; Radford et al., 2018) and can generate human-like text, enabling them to act as synthetic respondents in survey and experimental research. By providing LLMs with demographic characteristic prompts (e.g., age, gender, socioeconomic status), researchers can elicit responses as though the model were a member of the specified group (Figure 8) (Kalinin, 2023; Weidinger et al., 2021).

The main advantages of using LLMs as synthetic respondents lie in their speed, cost-effectiveness, and scalability. LLMs can quickly generate large volumes of data at a fraction of the cost compared to traditional surveys, making them useful in exploratory research (Amirova et al., 2023; Møller et al., 2023) or studying sensitive topics (Hirschberg & Manning, 2015), particularly for independent researchers or students lacking institutional resources to collect data from human respondents. LLMs can also maintain consistency across responses for hypothesis testing that requires stable respondent characteristics. Early studies show that LLMs can replicate behavioral economics paradigms and political psychology experiments with reasonable accuracy, suggesting their potential as synthetic respondents in academic research (Ashokkumar et al., 2024; Horton, 2023). Within the industry, marketing professionals have found success employing LLMs to simulate qualitative focus groups for brand sentiment analysis (Vriens et al., 2024).

However, despite early evidence showing LLMs' use for generic experimentation, they are currently limited by *algorithmic fidelity* and *source bias* for more nuanced research inquiries. Algorithmic fidelity refers to how well LLMs replicate human beliefs and behaviors (Argyle et al., 2023). Although LLMs generate coherent responses, they lack real human experiences, emotions, and intent which limits their use in certain domains requiring such factors (Bender et al., 2021; Kalinin, 2023). For instance, LLMs trained on data relevant to the Gambler's Fallacy will show little to no susceptibility to it in contrast to human respondents (Falk & Konold, 1997; Wagenaar, 1972). Additionally, overfitting to prompt structures can result in homogeneous outputs, reducing the diversity of opinions within a demographic. LLMs also inherit biases from their training data, which can amplify stereotypes or distort findings when simulating marginalized or underrepresented groups (Sheng et al., 2019; Weidinger et al., 2021). In other words, LLMs perform better on well-represented topics, while nuanced or niche queries may yield poorer-quality responses. For instance, a recent analysis conducted by Kantar, a prominent market research firm, revealed that GPT-4-generated synthetic responses often reinforced stereotypes, displayed excessively positive consumer tendencies, and lacked nuanced understanding of demographic subgroups (Ostler & Kalidas, 2023). The opacity of LLMs' internal workings—the *black box problem*—further complicates efforts to attenuate these biases. Integrating SYDNI with synthetic respondent sampling could address some of these limitations. By introducing random noise into LLM-generated data, SYDNI can enhance heterogeneity, offering a way to partially 'unbias' datasets and provide more conservative effect size estimates.

6 Conclusion

Although data balancing techniques like SMOTE and GANs are standard tools among data scientists, particularly in industry applications for machine learning, their use for subgroup analyses and data augmentation remains relatively new. Only recently have SaaS (Software as a Service) and B2B (Business-to-Business) companies begun offering these data augmentation services to marketing vendors and clients. Even more so, their adoption in academia has been largely absent. A likely barrier to broader academic adoption is the potential for misuse, where tools like SYDNI could facilitate p-hacking if not carefully applied. However, this risk underscores a broader challenge within the social and business sciences: an overreliance on binary significance testing where arbitrary p-value cutoffs often take precedence over qualitatively evaluating practical or substantive significance through a holistic and topic-contextual interpretation of effect size, sample size, and measurement uncertainty.

As research methodologies in these fields evolve toward emphasizing qualitative and practical significance over traditional significance testing, SYDNI holds potential as a valuable tool for robustness testing. Furthermore, as advancements in AI continue to refine LLMs, the potential for combining SYDNI with LLM-generated synthetic respondents becomes increasingly promising. Such a hybrid model could simulate greater sample heterogeneity even with potential LLM biases, enabling more realistic and nuanced data augmentation. Importantly, these advancements may help make academic publishing more accessible by offering a means to conduct robustness checks without relying on extensive institutional resources—such as funding, respondent pools, and IRB infrastructure–that are often concentrated within certain scholarship circles. As more Ph.D. researchers enter industry due to a shrinking academic job market, a collaborative approach between academic and industry researchers could lead to open-access innovations that bridge these gaps, offering greater opportunity for a wider range of researchers to explore and publish meaningful insights into human behavior.

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Table A1: Direct Effects of Observed Consumers

	Model 1 Transcendence	Model 2 Openness	Model 3 Enhancement	Model 4 Conservation	Model 5 Brand	Model 6 Product
Variable	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*	β 95% *CI*
self	0.011	0.387 ***	0.486 ***	0.126 **	0.060	0.040
	(-0.063, 0.094)	(0.306, 0.463)	(0.409, 0.560)	(0.033, 0.211)	(-0.034, 0.147)	(-0.063, 0.136)
fulfillment	0.016	0.151 ***	-0.092 *	-0.046	-0.021	0.049
	(-0.057, 0.092)	(0.071, 0.231)	(-0.169, -0.010)	(-0.132, 0.037)	(-0.104, 0.070)	(-0.046, 0.146)
affection	0.482 ***	-0.044	-0.077	0.263 ***	0.096	0.031
	(0.407, 0.552)	(-0.143, 0.041)	(-0.163, 0.003)	(0.167, 0.349)	(-0.001, 0.192)	(-0.078, 0.132)
transcendence					-0.110 **	-0.057
					(-0.192, -0.026)	(-0.140, 0.021)
openness					0.300 ***	0.206 ***
					(0.219, 0.377)	(0.118, 0.293)
enhancement					-0.030	-0.027
					(-0.115, 0.055)	(-0.122, 0.068)
conservation					0.009	-0.015
					(-0.073, 0.085)	(-0.089, 0.059)
Observations	924	924	924	924	924	924
R^2	0.273	0.254	0.256	0.151	0.111	0.082
R^2 Adj	0.266	0.246	0.248	0.143	0.099	0.069
F	38.204	34.538	34.878	18.046	8.776	6.254

Note:
Demographic variables and other covariates not shown.

Appendix

Table A1 provides the PLS-SEM direct effects for the observed data with no SYDNI augments. Self-Enjoyment and Fulfillment were positive correlates of Openness. Affection was a strong positive correlate of Self-Transcendence. In turn, Openness was the biggest positive driver of both brand and product adoption while Self-Transcendence was a negative driver of brand adoption. Accordingly, Self-Transcendence mediated the link between Affection and Brand Adoption (β = -0.053, SE = 0.021, z = -2.551, 95% CI [-0.093, -0.012], p = 0.011). Openness likewise mediated the effects of Self-Enjoyment ($\beta = 0.116$, SE = 0.020, $z = 5.852$, 95% CI [0.080, 0.156], $p < 0.001$) and Fulfillment ($\beta = 0.045$, SE = 0.014, $z = 3.347$, 95% CI [0.021, 0.074], $p < 0.001$) on Brand Adoption. Similarly, Openness mediated the effects of Self-Enjoyment ($\beta = 0.080$, SE = 0.020, $z = 4.033$, 95% CI [0.043, 0.119], $p < 0.001$) and Fulfillment ($\beta = 0.031$, SE = 0.011, $z = 2.906$, 95% CI [0.013, 0.054], p = 0.004) on Product Adoption.

Table A2 displays the model effects for *observed* high-income respondents, while Table **??** presents the same effects for *SYDNI-augmented* high-income respondents. In the observed high-income sample, Affection had a strong effect on Transcendence, but this effect became unstable when minor noise was introduced via SYDNI augmentation, as indicated by a divergence in the p-value and 95% Confidence Interval[1]. This suggests that the effect of Affection on Transcendence may be vulnerable to effect fluctuations as sample heterogeneity increases. Similarly, the effect size

[1]Discrepancies between p-values and confidence intervals in bootstrapped tests stem from differences in their underlying interpretations. The p-value reflects the proportion of bootstrapped test statistics that are as extreme as, or more extreme than, the observed test statistic. In contrast, the confidence interval represents the range between the 2.5th and 97.5th percentiles of the bootstrapped estimates. When $p < 0.05$ but the confidence interval includes zero, it indicates potential instability or uncertainty in the effect estimate. This discrepancy suggests that the result may be sensitive to minor fluctuations in the data or that the bootstrapped distribution is skewed or asymmetric.

Table A2: Direct Effects of Observed High Income Consumers

	Model 1 Transcendence	Model 2 Openness	Model 3 Enhancement	Model 4 Conservation	Model 5 Brand	Model 6 Product
Variable	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*
self	-0.099	0.323 **	0.439 ***	0.158	0.217	0.316 *
	(-0.392, 0.205)	(0.059, 0.539)	(0.203, 0.666)	(-0.145, 0.409)	(-0.077, 0.494)	(0.077, 0.565)
fulfillment	0.155	-0.035	-0.063	-0.017	-0.094	-0.118
	(-0.183, 0.442)	(-0.299, 0.210)	(-0.331, 0.233)	(-0.340, 0.323)	(-0.381, 0.185)	(-0.388, 0.140)
affection	0.390 **	0.068	-0.010	0.163	-0.006	-0.095
	(0.102, 0.634)	(-0.175, 0.302)	(-0.236, 0.196)	(-0.170, 0.455)	(-0.254, 0.238)	(-0.395, 0.176)
transcendence					-0.065	-0.096
					(-0.322, 0.221)	(-0.333, 0.147)
openness					0.364 **	0.376 **
					(0.101, 0.631)	(0.123, 0.632)
enhancement					-0.063	-0.140
					(-0.340, 0.223)	(-0.410, 0.126)
conservation					-0.031	-0.019
					(-0.265, 0.215)	(-0.246, 0.208)
Observations	123	123	123	123	123	123
R^2	0.221	0.206	0.356	0.164	0.172	0.209
R^2 Adj	0.159	0.143	0.304	0.098	0.073	0.115
F	3.555	3.266	6.933	2.470	1.738	2.219

Note:
Demographic variables and other covariates not shown.

Table A3: Direct Effects of Augmented High Income Consumers

	Model 1 Transcendence	Model 2 Openness	Model 3 Enhancement	Model 4 Conservation	Model 5 Brand	Model 6 Product
Variable	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*	*β* 95% *CI*
self	-0.153	0.341 ***	0.334 ***	0.124	0.162	0.289 **
	(-0.375, 0.120)	(0.161, 0.506)	(0.139, 0.523)	(-0.089, 0.311)	(-0.049, 0.352)	(0.109, 0.484)
fulfillment	0.086	-0.033	-0.098	-0.120	-0.197	-0.115
	(-0.165, 0.301)	(-0.190, 0.131)	(-0.277, 0.090)	(-0.310, 0.070)	(-0.392, -0.009)	(-0.299, 0.050)
affection	0.419 *	0.056	0.036	0.178	0.105	-0.063
	(-0.246, 0.587)	(-0.105, 0.215)	(-0.107, 0.180)	(-0.031, 0.374)	(-0.077, 0.286)	(-0.253, 0.139)
transcendence					-0.148	-0.013
					(-0.348, 0.140)	(-0.313, 0.305)
openness					0.378 ***	0.227 *
					(0.197, 0.571)	(0.057, 0.404)
enhancement					-0.065	-0.034
					(-0.235, 0.103)	(-0.184, 0.114)
conservation					0.023	-0.016
					(-0.166, 0.219)	(-0.183, 0.153)
Observations	277	277	277	277	277	277
R^2	0.192	0.184	0.272	0.184	0.162	0.151
R^2 Adj	0.165	0.157	0.247	0.157	0.121	0.109
F	7.060	6.711	11.074	6.705	3.924	3.602

Note:
Demographic variables and other covariates not shown.

of Openness on Product Adoption diminished in the SYDNI-augmented sample, further indicating the sensitivity of this effect to sample heterogeneity, albeit the SYDNI sample effect is closer to the observed global effect. The increased variability in effect sizes introduced by SYDNI highlights potential instability in the observed data, warranting cautious interpretation of these findings and the benefits of robustness checks in social and business sciences using data augmentation techniques.